REGRESSION OF UVEAL MELANOMA AFTER RU-106 BRACHYTHERAPY AND THERMOTHERAPY BASED ON METABOLIC ACTIVITY MEASURED BY POSITRON EMISSION TOMOGRAPHY/ COMPUTED TOMOGRAPHY

CHRISTOPHER S. LEE, MD,* SUNG C. LEE, MD,* KYUNGSIK LEE, MD,* HEE J. KWON, MD,* JEONG H. YI, MD,* ARTHUR CHO, MD†

Purpose: To evaluate regression rates of uveal melanoma after combined Ru-106 plaque radiotherapy and thermotherapy according to metabolic activity measured by positron emission tomography/computed tomography imaging.

Methods: A retrospective medical chart review was conducted on 26 patients with uveal melanoma who underwent pretreatment whole-body positron emission tomography/ computed tomography and received combined plaque radiotherapy and thermotherapy between 2006 and 2011. Tumors were classified as metabolically active and inactive based on the positron emission tomography/computed tomography imaging and compared with tumor height regression rates after treatment.

Results: Before treatment, the median tumor thickness was 8.8 mm for metabolically active tumors (7 eyes) and 5.0 mm for metabolically inactive tumors (19 eyes). The median tumor thicknesses with respect to the original thickness at 3, 6, and 12 months after treatment were 88%, 78%, and 64% for metabolically active tumors and 95%, 89%, and 81% for metabolically inactive tumors, respectively. The monthly tumor regression rates during the first 3 months (4.2% vs. 1.7%, $P = 0.022$) and the overall monthly tumor regression rates (3.0% vs. 1.5%, $P = 0.041$) were significantly higher for metabolically active tumors versus inactive tumors. Two patients with positive metabolic activity developed metastatic diseases 2 years after treatment, whereas no patient with negative metabolic activity developed metastatic disease during the study period.

Conclusion: Positive metabolic activity of uveal melanoma based on the positron emission tomography/computed tomography was significantly associated with rapid initial tumor regression after combined plaque radiotherapy and thermotherapy, suggesting a prognostic value for this diagnostic approach.

RETINA 34:182–187, 2014

Plaque radiotherapy has been an important treatment of uveal melanoma, especially after the Collaborative Ocular Melanoma Study showed comparable survival for patients with medium-sized choroidal melanoma treated with plaque radiotherapy versus enucleation.[1,2] Somewhat counterintuitively, several studies have demonstrated that faster tumor regression in uveal melanoma after radiotherapy correlates with systemic metastases and mortality.[3–5] It was therefore speculated that tumors more active in metabolism and mitosis could be more responsive to radiotherapy. Supporting this finding, radiosensitive melanomas after Ru-106 brachytherapy were associated with epithelioid cells.[6] Mixed cell melanomas exhibited faster growth and higher mitotic activity compared with spindle cell melanomas.[7] In addition, faster growing tumors regress more rapidly after plaque radiotherapy than slower growing tumors,[8] and larger tumors were reported to shrink faster than smaller tumors after plaque radiotherapy.[9] Monosomy 3, a strong predictor of poor survival for uveal melanoma, has been associated with faster tumor regression after plaque radiotherapy.[10]

Recently, our group demonstrated that high metabolic activity measured by positron emission tomography (PET) imaging at presentation is associated with poor survival for patients with uveal melanoma.[11] Metabolically active tumors show greater F-18 fluorodeoxyglucose (FDG) uptake during PET imaging, which is represented as standardized uptake values (SUV). High SUV was also shown to be associated with poor prognostic factors for uveal melanoma, including monosomy 3,[12–14] and tumors with high SUV metastasizing earlier.[11] These observations led us to hypothesize that metabolically active tumors based on the PET imaging are associated with rapid tumor regression after plaque radiotherapy.

Methods

Patients

This study was approved by the Institutional Review Board of Yonsei University College of Medicine. The medical records of all patients diagnosed with uveal melanoma at the Department of Ophthalmology, Yonsei University College of Medicine, Seoul, Korea, between October 2006 and December 2011 were reviewed. Twenty-six patients with uveal melanoma who had undergone a pretreatment PET/computed tomography (CT) received combined brachytherapy using Ru-106 plaque (BEBIG, Berlin, Germany) and thermotherapy. The target radiation dose was 85 Gy to the tumor apex with a maximum of 1,000 Gy to the sclera. All patients with brachytherapy received adjunctive transpupillary thermotherapy (TTT) (Iris Medical, Mountain View, CA) of the tumor apex before plaque insertion in the operating room, using indirect ophthalmoscope. The second and third planned sessions of TTT were provided in the office setting using a slit-lamp delivery system at 3-month intervals. The fourth session of TTT was considered for cases who were deemed to have insufficient tumor regression. Exposure time was 1 minute per application. Laser energy (300–1200 mW) was adjusted stepwise until the tumor surface turned slightly gray at the end of the 1-minute exposure. The largest basal tumor diameter and tumor height were measured by B-scan echography. Chromosome 3 status (monosomy 3 or disomy 3) was not included in the analysis because of limited data.

Whole-Body Dual-Modality Positron Emission Tomography/Computed Tomography

All patients underwent routine FDG PET/CT with DSTe PET/CT (GE Healthcare, Milwaukee, WI). All patients fasted for at least 6 hours, and glucose levels in the peripheral blood were confirmed to be 140 mg/dL or less before FDG injection. Approximately 5.5 MBq/kg of body weight of FDG were administered intravenously 1 hour before image acquisition. After the initial low-dose CT (DSTe: 30 mAs, 130 kVp), a standard PET imaging protocol from the neck to the proximal thighs with an acquisition time of 3 minutes per bed in three-dimensional mode was taken. Images were then reconstructed using the ordered subset expectation maximization (2 iterations, 20 subsets).

Imaging Analysis

For evaluation of the [^{18}F] FDG uptake of the ocular lesion at the initial PET/CT before operation, an experienced nuclear medicine specialist (A.C.) interpreted the PET images by maximum SUV. A region of interest was drawn at the ocular lesion seen in the corresponding PET/CT, and maximum SUV was measured. Values $\geq$2.2 SUV were indicative of positive metabolic activity as determined in a previous study.[11]

Statistical Analysis

The Mann–Whitney U test, Fisher exact test, and log-rank test were used to compare demographics, tumor characteristics, and tumor regression rates between metabolically active tumors and metabolically inactive tumors. Statistical analysis was performed using SPSS 15.0 (SPSS, Inc, Chicago, IL), and a P value $<$0.05 was considered significant for all analyzes.

Results

Twenty-six eyes of 26 patients were included in this study. Seven patients (27%) were positive for metabolic activity by PET/CT, and 19 patients (73%) were negative for metabolic activity. The median age was

From the *Department of Ophthalmology, The Institute of Vision Research and †Department of Nuclear Medicine, Yonsei University College of Medicine, Seoul, Korea.

Supported by a grant form the Korean Health Technology R&D Project, Ministry for Health, Welfare & Family Affairs, Korea (A110272).

Presented in part at the Asia-Pacific Academy of Ophthalmology Congress, Busan, Korea, April 13–16, 2012, and at the Association for Research in Vision and Ophthalmology Annual Meeting, Fort Lauderdale, FL, May 6–10, 2012.

None of the authors have any conflicting interests to disclose.

Reprint requests: Arthur Cho, MD, Division of Nuclear Medicine, Department of Diagnostic Radiology, Yonsei University College of Medicine, Seoul, Korea; e-mail: artycho@yuhs.ac

54 years (range, 25–82 years), and 15 patients (58%) were men. The median follow-up was 13 months (range, 3–43 months). The median largest tumor basal diameter was 10.9 mm (range, 6.3–17.3 mm), and the median tumor height was 5.4 mm (3.1–10.2 mm). Eighteen patients (69%) underwent 3 planned sessions of adjuvant TTT. Two patients with metabolically active tumor and three patients with metabolically inactive tumors underwent the fourth TTT. One patient with metabolically active tumor and another patient with metabolically inactive tumor did not undergo additional TTT sessions after the first intraoperative TTT because of significant tumor regression. One patient with metabolically active tumors did not receive the third planned TTT session because of follow-up loss. There was no statistical difference in the median number of TTT sessions or total TTT time between metabolically active tumors and metabolically inactive tumors. Demographics and tumor characteristics based on the metabolic activity by PET/CT are listed in Table 1.

At presentation, the median tumor thickness by ultrasonography for metabolically active tumors was 8.8 mm compared with 5.0 mm for metabolically inactive tumors. The median tumor thicknesses with respect to original thickness at 3, 6, and 12 months after treatment were 88%, 78%, and 64% for metabolically active tumors and 95%, 89%, and 81% for metabolically inactive tumors, respectively (Figure 1). Percent decrease in tumor thickness at 3 months (88% vs. 95%, P = 0.022) and the monthly rate of decrease in tumor thickness at 3 months (4.2% vs. 1.7%, P = 0.022) were significantly greater in metabolically active tumors compared with metabolically inactive tumors (Table 2). Percent decrease in tumor thickness at 6 and 12 months was not significantly different between the groups. The overall monthly median tumor regression rate was greater for metabolically active tumors versus inactive tumors (3.0% vs. 1.5%, P = 0.041).

Linear regression analysis showed that initial tumor height did not correlate with tumor height regression at 3 months (P = 0.864), 6 months (P = 0.779), and 12 months (P = 0.900) among metabolically active tumors, whereas it correlated among metabolically inactive tumors (P = 0.018, P = 0.028, P = 0.015, respectively).

Two patients (29%) with metabolically active tumors developed metastatic disease (liver metastases) 2 years after treatment, whereas no patients with

Table 1. Demographics and Tumor Characteristics of Uveal Melanoma Based on Metabolic Activity by PET/CT

Features	Metabolically Active Tumors	Metabolically Inactive Tumors	P*
No. of eyes	7	19	
Age, median (range), years	52 (25–69)	55 (26–82)	0.572
Male/female	3/4	12/7	0.665†
Right/left	2/5	8/11	0.668†
Tumor size			
LBD, median (range), mm	13.1 (10.9–17.3)	9.9 (6.3–15.4)	0.010
Height, median (range), mm	8.8 (5.3–10.2)	5.0 (3.1–9.8)	0.004
According to TNM (7th edition)‡			0.051†
T1	0 (0%)	7 (37%)	
T2	2 (29%)	8 (42%)	
T3	5 (71%)	4 (21%)	
T4	0 (0%)	0 (0%)	
Anterior margin of tumor			0.999†
Anterior to ora serrata	0 (0%)	1 (5%)	
Ora serrata to equator	2 (29%)	5 (26%)	
Posterior to equator	5 (71%)	13 (69%)	
Posterior margin of tumor			0.452†
<2 DD from fovea	4 (57%)	5 (26%)	
≥2 DD from fovea	2 (29%)	10 (53%)	
<2 DD from optic disk	1 (14%)	4 (21%)	
SUV, median (range)	2.6 (2.4–5.1)	1.5 (0.7–1.9)	<0.001
Metastatic disease	2 (29%)	0 (0%)	0.013§
No. of TTT, median (range)	3 (1–4)	3 (1–4)	0.778
Total TTT time, median (range), minutes	42 (15–61)	40 (12–50)	0.306
Apex dose, Gy (range)	35 (20–85)	85 (25–101)	0.012
Scleral contact dose, Gy (range)	835 (650–1136)	645 (320–1177)	0.097

*Mann–Whitney U test.
†Fisher exact test.
‡AJCC/UICC (American Joint Cancer Committee/Union Internationale Contre le Cancer).
§Log-rank test.
DD, disk diameter; LBD, largest basal diameter.

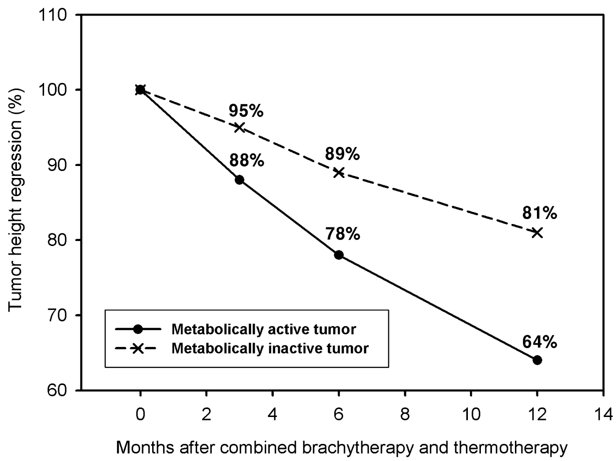

Fig. 1. Tumor height regression as a percentage of initial tumor height after combined Ru-106 plaque radiotherapy and thermotherapy based on the metabolic activity using PET/CT at presentation.

metabolically inactive tumors developed metastatic disease during the study period.

Discussion

Radiation can kill tumor cells either directly or indirectly. The indirect effect is believed to be a more important mechanism and includes the formation of free radicals in the cells through photon ionization. These free radicals damage DNA or other vital components of cells, leading to cell death.[15] Radiosensitivity of a cell can be 4-fold greater during the mitotic phase than during the interphase.[16] Thus, it is possible that more rapidly dividing and highly metabolic tumors would be more responsive to radiotherapy and do worse.

Indeed, several studies have correlated rapid tumor regression in uveal melanoma after radiotherapy as a poor prognostic factor. Cruess et al[17] first showed that the extent and rate of tumor regression in metastatic melanoma after Cobalt-60 plaque radiotherapy was not appreciably different than the rate and extent of tumor regression in non-metastatic patients, making it a poor indicator of prognosis. However, the same group later found that tumor regression at 1 year after treatment had an unfavorable prognostic value.[3] Glynn et al[4] found that patients with rapid tumor regression were more likely to metastasize in the first 2 years after proton beam irradiation compared with those with slow tumor regression. Kaiserman et al[5] showed that the initial rate (during the first 3 months) of tumor regression after Ru-106 brachytherapy had a prognostic value. Finally, rapid tumor regression has been associated with known poor prognostic factors, such as larger tumor size[9] and monosomy 3.[10]

There has been recent interest in PET imaging to characterize initial staging, recurrence, and metastasis of uveal melanomas.[11,18–23] Combining PET with CT has increased the diagnostic accuracy compared with PET imaging alone.[24] Finger and Chin[25] have demonstrated that 16 of 18 (89%) uveal melanomas with SUV greater than 2.0 resulted in undetectable SUV after plaque brachytherapy during the median follow-up of 16 months, suggesting the usefulness of PET imaging in the assessment of tumor metabolism and treatment effects. High SUV, indicative of high tumor metabolic activity, has been shown to be associated with known risk factors for metastasis in uveal melanoma, such as increasing tumor size, epithelioid cell type, and monosomy 3.[12,14,19,21] High SUV is also correlated with early metastasis and poor patient survival.[11] As anticipated, tumors with high SUV showed rapid tumor regression in this study. Faster monthly tumor regression rates were evident at all time intervals, but statistical significance was only observed at 3 months (88% of initial height or 4.2% per month versus 95% or 1.7% per month). This could suggest that metabolic activity by PET/CT is more correlated with immediate tumor response to radiotherapy or could simply mean the decreased statistical power with the decreasing number of patients with longer follow-up periods. Interestingly, after treatment, some metabolically inactive tumors grew taller, whereas all metabolically active tumors regressed (Table 2). Overall, the monthly tumor regression rate was also statistically greater for metabolically active tumors versus inactive tumors (3.0% vs. 1.5%).

Table 2. The Median Monthly Rate of Decrease in Tumor Thickness After Combined Brachytherapy and Thermotherapy at Each Time Intervals Based on Metabolic Activity by PET/CT (%/Month)

Time Intervals (months)	Metabolically Active Tumors	Metabolically Inactive Tumors	Total	*P**
Overall	3.0 (1.1 to 5.8) (n = 7)	1.5 (−2.5 to 7.1) (n = 19)	2.1 (n = 26)	0.041
0–3	4.2 (2.3 to 15.1) (n = 7)	1.7 (−7.5 to 20.5) (n = 19)	2.5 (n = 26)	0.022
3–6	3.4 (0.6 to 7.1) (n = 5)	1.3 (−8.2 to 20.0) (n = 18)	1.4 (n = 23)	0.491
6–12	2.4 (0 to 6.4) (n = 5)	1.0 (−1.7 to 3.3) (n = 13)	1.1 (n = 18)	0.566
0–6	3.6 (2.3 to 8.5) (n = 5)	1.8 (−4.9 to 14.1) (n = 18)	2.3 (n = 23)	0.150
0–12	3.0 (1.1 to 5.8) (n = 5)	1.6 (−1.0 to 4.3) (n = 13)	2.1 (n = 18)	0.173

*Mann–Whitney *U* test between metabolically active tumors and inactive tumors.

The initial rate of tumor height regression after Ru-106 brachytherapy (during ~24 months) is reported to be ~3% per month.[9] After Ru-106 brachytherapy, tumors usually do not regress completely, but their heights are reported to be stabilized at ~60% to 70% of the initial height after ~24 months to 36 months.[9,26] Kaiserman et al[9] demonstrated that large tumors (height >8 mm) show faster initial decrease in height and stabilize at ~50% of their initial height after Ru-106 brachytherapy compared with small tumors (height <4 mm) that stabilize at ~70%. Although larger tumors generally received more radiation to the middle and the base of the tumor because 100 Gy was targeted to the tumor apex for all cases, Kaiserman et al[9] reasoned that different tumor responses to radiation are not due to the different radiation dose tumors received because 97% of tumors shrunk in size and did not recur. Both small and large tumors received sufficient radiation to kill proliferating cells, and further radiation would not be expected to result in further tumor regression in a dose-dependent manner. Tumor radiosensitivity is related to tumor metabolism, so large tumors are expected to show higher SUV.[19,21] Indeed, metabolically active tumors showed a greater initial basal diameter and height compared with metabolically inactive tumors in this study. However, tumor size may not be the only factor that determines SUV positivity; all 26 tumors had basal diameter >6 mm, being higher than the minimal resolution of our PET/CT, but only 7 tumors (27%) showed positive metabolic activity. Positive SUV rate in previous studies ranged from 17% to 60%.[14,19,21,25] Among metabolically active tumors, initial tumor height was not correlated with tumor regression after treatment at all time intervals. Initial tumor height, however, was correlated with tumor regression among metabolically inactive tumors. This finding may imply that positive SUV can have a prognostic value in postbrachytherapy response independent of initial tumor height.

We have used a combination of TTT with plaque radiotherapy to secure better local tumor control, especially when tumors are located near the optic disk and fovea. Transpupillary thermotherapy can result in tumor necrosis up to 3.9 mm deep and occlusion of blood vessels in the treated area.[27] The maximum tumor height adequate for Ru-106 brachytherapy appears to be 5 mm to 7 mm.[27,28] Combining both treatments, which was called "sandwich therapy" first by Journee-de Korver et al,[29] makes it possible to treat tumors thicker than 5 mm to 7 mm. In addition, thermotherapy can effectively kill and radiosensitize tumor cells that are relatively radioresistant, including S-phase cells, nutrient-deficient cells, and cells at a low pH.[30] These synergistic effects have led us to consider sandwich therapy for all tumors, even those that are thinner than 5 mm. Adjuvant TTT is a confounder in assessing the radiosensitivity and tumor response to radiotherapy in this study. However, there was no difference in the number of TTT sessions or total TTT times between metabolically active and inactive tumors, mainly because our protocols included 3 planned TTT sessions at 3-month interval (Table 1). Moreover, a linear regression test showed that there was no correlation between the number of TTT and regression of tumor height (data not shown). So, we think the difference in the rate of tumor regression between metabolically active and inactive tumors is not attributed to the adjuvant TTT in our patients.

The overall monthly tumor regression rate was 2.1%. This tumor regression rate could be somewhat lower than that of the previously mentioned studies. Uveal melanomas usually regress faster in initial period after radiotherapy, thus showing an exponential decay curve.[9] Considering that the study period for tumor regression in this study was 1 year after treatment, tumor regression rates during the initial year would be expected to be somewhat higher, if not comparable, to what has been previously reported with longer follow-ups. This difference could be attributed to the presence of more patients with metabolically inactive tumors in this study cohort. This study is limited by retrospective design and few patients, which can affect the statistical significant of our findings. Further long-term studies with more patients are therefore necessary to better evaluate tumor response to radiotherapy in this population.

In summary, we have shown that metabolically active uveal melanomas as assessed by PET/CT were associated with rapid initial tumor height regression. Positron emission tomography/CT may therefore be useful in predicting tumor response to radiotherapy, in determining likely prognosis, and in deciding follow-up regimens after treatment.

Key words: PET/CT, prognosis, tumor regression, uveal melanoma.

Acknowledgments

We thank Dr. Hyung Tak Lim for his support in statistical analysis.

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